

PRISM: Training-Free Video Anomaly Detection via Intrinsic Statistical Modeling

Yuantong Chen¹ Zhengyan Ding¹ Yanfeng Shang¹

Abstract

While recent training-free video anomaly detection (VAD) methods offer advantages such as interpretability and ease of deployment, they often suffer from computational inefficiency due to complex memory retrieval mechanisms or high-latency visual-language models (VLMs). To address this issue, we propose PRISM (Parameterless Recognition Based on Intrinsic Statistical Modeling), a novel framework for efficient open-set anomaly detection with minimal computational cost. Built on a pre-trained multimodal embedding model, PRISM introduces differential amplification and whitening mechanisms to statistically suppress common-mode background noise in the embedding space, thereby improving the signal-to-noise ratio of anomalous events. Extensive experiments on three widely used datasets demonstrate that PRISM achieves state-of-the-art performance among training-free methods while maintaining real-time inference capability. Furthermore, our statistical analysis offers a complementary perspective on why training-free methods may suffer from lower Average Precision (AP) on complex datasets such as XD-Violence. Code is released at <https://github.com/ytC2026/ICML2026-PRISM>.

1. Introduction

With the explosive growth in the number of videos worldwide, relying solely on manual processing of massive amounts of video data is no longer feasible. Therefore, Video anomaly detection (VAD) is an important task for automatically identifying abnormal events from large-scale video streams. VAD (Video Anomaly Detection) aims to

¹The Third Research Institute of Ministry of Public Security, Shanghai, China. Correspondence to: Zhengyan Ding <dingzhengyan@gass.ac.cn>.

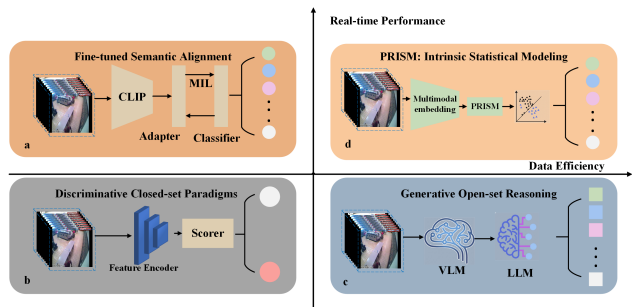


Figure 1. Comparison of VAD paradigms. Existing research can be categorized into three types: **a) Fine-tuned Semantic Alignment**, **b) Discriminative Closed-set Paradigms**, and **c) Generative Open-set Reasoning**. To address the limitations of training costs and inference latency in these approaches, our proposed framework is presented as: **d) PRISM**, which utilizes **Intrinsic Statistical Modeling** of video text vector for robust, training-free detection.

significantly improve system safety and efficiency in fields such as healthcare, industry, and autonomous driving by automatically identifying abnormal events (such as violence and accidents) through the analysis of video data, thereby alleviating the burden on human labor.

Early research on video anomaly detection fell mainly into two categories. One was a self-supervised paradigm based on reconstruction (Hasan et al., 2016; Gong et al., 2019) or prediction (Liu et al., 2018; Park et al., 2020); The other was a weakly supervised paradigm based on feature discrimination, which typically used pre-trained networks such as I3D or C3D ((Wan et al., 2021)) to extract high-level semantic features and then fed them into a multiple instance learning (MIL) framework for anomaly score prediction. Although these methods have made significant progress on specific datasets, they still face core challenges in generalization and interpretability in practical deployments.

In recent years, with the rapid development of Vision-Language Pre-training (VLPs), many researchers have made good use of the semantic representation capabilities of these models. CLIP-based Task-specific models have achieved remarkable results and breakthroughs in various recognition tasks (Wu et al., 2023) (Wu et al., 2024). However, the actual deployment cost remains a limiting factor because they still require full training and testing.

Recently, with the rapid development of Visual-Language Models (VLMs), a series of zero-shot VAD methods based on large models have emerged, providing new ideas for the field (Li et al., 2025)(Ye et al., 2025)(Zanella et al., 2024)). For example, some researchers have used VLMs to generate frame-level captions and combined them with various prior knowledge and structured designs, using LLMs for inference to achieve truly training-free detection. However, these generative model-based methods are often accompanied by huge inference latency (inference speed of only about 1 FPS) and rely on complex prompt engineering, making it difficult to meet real-time monitoring requirements.

To address these issues, model-based video embedding methods, which have emerged in recent years, have become a promising solution. Previous researchers directly mapped video clips and text descriptions to the same feature space, achieving efficient real-time detection through a retrieval mechanism. However, direct retrieval still faces two key problems (Lee et al., 2025): (1) requiring extensive knowledge bases with high deployment costs, and (2) suffering from "anomaly bias"—normal backgrounds showing high similarity to anomalous text. To mitigate this bias, previous researchers employed a heuristic "scaled penalization" strategy. Although effective, this approach is essentially an engineering compromise and lacks an explicit modeling of feature-space geometry.

We argue that this anomaly bias is closely related to common-mode noise in the feature space (redundant environmental features shared by both normal and anomalous events), which affect the model's judgment. By leveraging the mechanism of differential amplifiers to suppress such common signals and combining classical machine learning theory with relevant insights from natural language processing, we propose a training-free "PRISM (Parameter-less Recognition Based on Intrinsic Statistical Modeling)" framework. Unlike previous scalar penalties, we vector a small amount of text generated by VLM to construct a covariance matrix, mathematically modeling and "whitening" away background noise. Experiments show that this method not only significantly enhances real-time and zero-shot models' advantages but also achieves significant improvements in signal-to-noise ratio (SNR), AP, and AUC. This results in performance that surpasses all current training-free models and approaches most current weakly supervised models. Furthermore, this perspective provides a possible explanation for the lower AP observed in training-free methods on XD-Violence.

Our main contributions are summarized as follows.

- We propose PRISM (Parameter-less Recognition based on Intrinsic Statistical Modeling), a novel training-free VAD framework. Unlike generative methods that suffer from high latency or traditional retrieval methods,

PRISM achieves real-time detection without a complex retrieval mechanism or heavy computational burden.

- We provide a new theoretical perspective on "anomaly bias," identifying it as "common-mode noise" in the feature space. Instead of using heuristic scalar penalization, we introduce a mathematical "whitening" mechanism. By constructing a covariance matrix from a small set of VLM-generated text vectors, we effectively model and eliminate background noise, significantly enhancing the Signal-to-Noise Ratio (SNR).
- Our method achieves state-of-the-art performance among training-free models and approaches weakly supervised benchmarks. Experiments show significant improvements in AP and AUC.

2. Related Work

2.1. Traditional and Training-based VAD

Early unsupervised VAD methods learn normal distributions via reconstruction (Hasan et al., 2016; Gong et al., 2019) or future frame prediction (Liu et al., 2018; Park et al., 2020). Weakly supervised approaches based on Multiple Instance Learning (MIL) extract spatiotemporal features (C3D, I3D) and optimize anomaly scores with Ranking Loss under video-level labels (Wan et al., 2021).

With the development of Vision-Language Pre-training (VLP), researchers have begun to use the semantic capabilities of CLIP into VAD. Works such as (Wu et al., 2023; 2024; Joo et al., 2023) no longer perform full fine-tuning on the pre-trained model, but instead freeze the CLIP image encoder and align video features with text semantics by designing learnable prompts or temporal adapters. However, these methods remain training-dependent on specific annotations, limiting zero-shot generalization to unseen scenarios.

2.2. Generative VLM-based VAD

To break free from dependence on training data, researchers began exploring the use of the general reasoning capabilities of large language models (LLM) and large visual models (VLM) to achieve zero-shot detection. The current methods can be broadly divided into two categories. The first category is represented by instruction-based fine-tuning methods, which typically use lightweight, deep neural network-based detectors to filter potentially anomalous frames. These frames are then fed into a Video Language Model (VLM) with specific cues for detailed description and judgment. The second category of methods includes training-free methods that rely on frozen models. These methods employ a "description and inference" process: first, the video is segmented using a fixed-length sliding time window, and then the VLM generates a description for each

segment, enabling the LLM to aggregate semantic dynamics and infer anomalies. While generative methods offer excellent interpretability and semantic understanding, they face significant challenges in practical deployment. First, the generative model results in extremely high inference latency (typically only 0.5-1 FPS), which cannot meet the real-time requirements of monitoring scenarios. Secondly, their performance is highly dependent on complex prompt and structural design, and they are prone to hallucinations, leading to misjudgments of normal behavior.

2.3. Video-Text Embedding and Feature Geometry

Unlike generative models, video embedding models directly map video and text to the same space through an encoder, achieving efficient retrieval-based detection by calculating cosine similarity. Previous researchers (Lee et al., 2025) have already achieved a very good result.

However, directly utilizing the original embedding vectors faces some problems. Existing research shows that the embedding space of pre-trained models generally suffers from representation degeneration, or anisotropy.

Specifically, all embedding vectors tend to be squeezed into a narrow cone, resulting in even semantically unrelated vector pairs (such as normal video and abnormal text) exhibiting high cosine similarity. Previous research attempts to mitigate this problem through scalar penalization, but this approach fails to fundamentally correct the distribution of the feature space mathematically. This paper addresses representation degradation through statistical whitening and manifold alignment. Unlike traditional scalar scaling, PRISM utilizes a precision matrix to eliminate inter-dimensional correlations, mathematically correcting for anisotropic spatial distributions. PRISM does not require a complex and massive database: we use only a very small and general set of text description (e.g., 280 sentences for **UCF-Crime**, 500 for **MSAD**, and 480 for **XD-Violence**) to define anomalous events and normal scenarios. During inference, we only need to match the most similar semantic axis for each vector and use semantics as the basis for interpretation, thereby maximizing the real-time performance of inference while avoiding calls to large online language models.

3. Approach

3.1. Analysis of Bias in Zero-Shot Detection

Since all embeddings are L2-normalized before similarity computation, cosine similarity is algebraically equivalent to the inner product:

$$\text{Sim}(\mathbf{a}, \mathbf{b}) = \mathbf{a}^\top \mathbf{b}. \quad (1)$$

However, in anisotropic embedding spaces (Gao et al.,

2019) (Schakel & Wilson, 2015) (Liang et al., 2021) (Yang et al., 2018), the dominant shared orientation can influence the inner product. This can lead to semantically mismatched video-text pairs, such as normal video clips and anomaly-related text descriptions, being assigned falsely high similarity scores.

To explain the systematic false positives in zero-shot detection from a mathematical perspective, we define a feature space $\mathcal{X} \subset \mathbb{R}^d$. For a given normal video segment feature $\mathbf{v}_n$, an ideal anomaly detector should satisfy the discrimination inequality Eq. (2).

$$\Delta S = \text{Sim}(\mathbf{v}_n, \mathbf{t}_a) - \text{Sim}(\mathbf{v}_n, \mathbf{t}_n) < 0 \quad (2)$$

where $\mathbf{v}_n$ denotes the normal video embedding, $\mathbf{t}_a$ and $\mathbf{t}_n$ represent the anomaly and normal text embeddings. Respectively, and $\text{Sim}(\cdot, \cdot)$ computes the cosine similarity. Note that the condition $\Delta S > 0$ indicates a false positive.

Since the current vector space exhibits significant anisotropy (i.e., the vector space is cone-shaped), we can assume that in the multimodal embedding model, the text-video vectors possess a principal axis of a hypercone. This direction represents the background direction (common-mode noise) for the vast majority of vectors. Therefore, generalizing its theory to the VAD field, we can assume that any multimodal embedding vector $\mathbf{x}$ can be orthogonally decomposed into an independent event semantic component $\mathbf{x}_e$ (Perpendicular to the principal axis of the cone) and a common-mode background component $\mathbf{x}_b$ (Cone center direction):

$$\mathbf{x} = \mathbf{x}_e + \mathbf{x}_b, \quad \text{s.t. } \mathbf{x}_e \perp \mathbf{x}_b \quad (3)$$

Since the pre-trained model enforces a normalization constraint $\|\mathbf{x}\| = 1$, the norms of these components satisfy the following relationship:

$$|\mathbf{x}_e|^2 + |\mathbf{x}_b|^2 = 1 \implies |\mathbf{x}_b| = \sqrt{1 - |\mathbf{x}_e|^2} \quad (4)$$

Let $\text{Freq}(\cdot)$ denote the occurrence frequency of the event concept. We leverage the statistical observation that vector magnitude serves as a measure of significance and is positively correlated with frequency in consistent contexts (Draganov et al., 2025) (i.e., $\|\mathbf{x}_e\| \propto \text{Freq}(\mathbf{x})^\gamma$ with $\gamma > 0$).

Since normal events $\mathbf{t}_n$ correspond to frequent, consistent patterns while anomaly descriptions $\mathbf{t}_a$ refer to rare events, we have the inequality $\text{Freq}(\mathbf{t}_n) > \text{Freq}(\mathbf{t}_a)$. Consequently, the norm of the semantic component satisfies:

$$\|\mathbf{t}_{n,e}\| > \|\mathbf{t}_{a,e}\| \quad (5)$$

By invoking the normalization constraint in Eq. (4), the inequality reverses for the complementary background components:

$$\|\mathbf{t}_{a,b}\| > \|\mathbf{t}_{n,b}\| \quad (6)$$

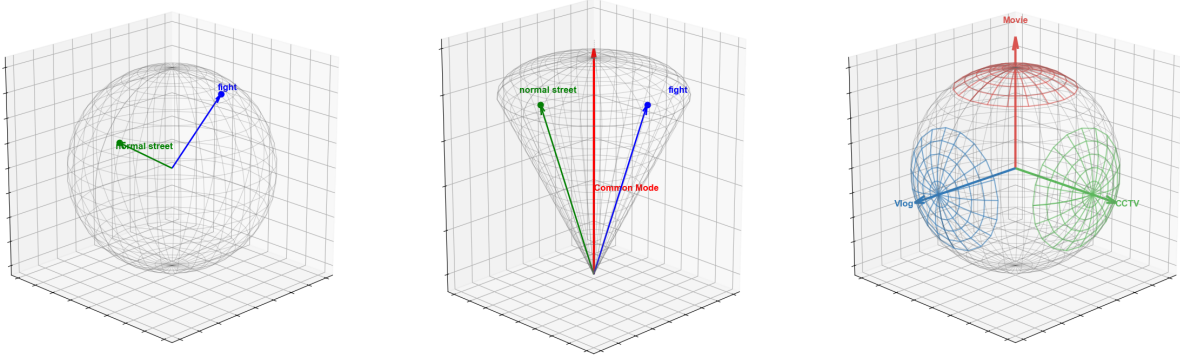


Figure 2. Comparison of embedding space geometries. Left: Ideal isotropic space. Middle: Anisotropic cone space showing the origin of systematic bias. Right: Ideal case after feature decoupling

Based on the derived norm inequality, we now investigate the mechanism of systematic false positives by substituting the vector decomposition into the discriminant function ΔS . This analysis relies on two geometric properties: (1) the orthogonality of mismatched semantics, where the normal video event is not correlated with the anomaly text (i.e. $\mathbf{v}_{n,e}^\top \mathbf{t}_{a,e} \approx 0$); and (2) the directional alignment of common-mode backgrounds due to spatial anisotropy (i.e. $\mathbf{v}_{n,b}^\top \mathbf{t}_b \approx \|\mathbf{v}_{n,b}\| \|\mathbf{t}_b\|$). Under these conditions, the discriminant function can be expanded and approximated as:

$$\Delta S \approx \underbrace{\mathbf{v}_{n,b}^\top (\mathbf{t}_{a,b} - \mathbf{t}_{n,b})}_{\text{Systematic Bias}} + \underbrace{(\mathbf{v}_{n,e}^\top \mathbf{t}_{a,e} - \mathbf{v}_{n,e}^\top \mathbf{t}_{n,e})}_{\text{Ideal Semantic Score}} \quad (7)$$

We identify the first term as the **Bias Term**. Since the background vectors are directionally aligned and the anomaly background norm is strictly larger than the normal one ($\|\mathbf{t}_{a,b}\| > \|\mathbf{t}_{n,b}\|$), this bias term remains strictly positive. In scenarios with strong background interference (i.e., large $\|\mathbf{v}_{n,b}\|$), the magnitude of this positive bias overwhelms the negative semantic term. Consequently, this leads to a false positive:

$$\begin{aligned} \mathbf{v}_{n,b}^\top (\mathbf{t}_{a,b} - \mathbf{t}_{n,b}) &> |\mathbf{v}_{n,e}^\top \mathbf{t}_{a,e} - \mathbf{v}_{n,e}^\top \mathbf{t}_{n,e}| \\ &\downarrow \\ \Delta S &> 0 \end{aligned} \quad (8)$$

The above analysis shows that directly using cosine similarity does not get good results. Therefore, a linear decorrelation transformation is strictly required to effectively suppress the common-mode background component (Systematic Bias). In section 3.2, we will introduce statistical whitening to suppress this common-mode noise.

3.2. Whitening Transformation

Following the analysis in Section 3.1, the anisotropic distribution of embeddings introduces a significant common-mode bias: generic scene and background semantics may occupy dominant directions in the embedding space, making standard cosine matching less sensitive to anomaly-specific signals. To mitigate this effect, we introduce a statistical whitening transformation inspired by classical machine learning theory. By whitening the text-side semantic anchors, PRISM decorrelates their second-order statistics and constructs anomaly axes that are less dominated by common semantic directions.

We therefore use a regularized covariance estimator:

$$\hat{\Sigma}_T = \Sigma_T + \lambda I, \quad (9)$$

where λ is a small regularization hyperparameter (e.g., $\lambda = 0.01$). Following standard ZCA whitening, we derive the whitening matrix as

$$\mathbf{W} = \hat{\Sigma}_T^{-1/2}, \quad (10)$$

such that the transformed text covariance is approximately isotropic. This operation reduces the influence of dominant shared directions in the text embedding space. As a result, anomaly axes are constructed with less interference from common semantic components.

Inference via Whitened Semantic Alignment. PRISM first builds the whitening matrix from the text description pool. All normal and anomaly text embeddings are L2-normalized. Let T_n denote the normal text set and $\{T_a^j\}_{j=1}^M$ denote the anomaly text groups. We compute the normal semantic center and form the full text pool:

$$\mu_n = \text{Mean}(T_n), T = T_n \cup T_a^1 \cup \dots \cup T_a^M. \quad (11)$$

Algorithm 1 PRISM Inference Pipeline

Input: video embeddings X_v , normal texts T_n , anomaly text groups $\{T_a^j\}_{j=1}^M$
Hyperparameters: $\lambda, \tau_{\text{sem}}, \sigma$
 Normalize all text embeddings in T_n and $\{T_a^j\}_{j=1}^M$.
 $\mu_n = \text{Mean}(T_n)$, $T = T_n \cup T_a^1 \cup \dots \cup T_a^M$
 $\Sigma_T = \text{Cov}(T)$, $\mathbf{W} = (\Sigma_T + \lambda I)^{-1/2}$
for $j = 1$ **to** M **do**
 $\mu_a^j = \text{Mean}(T_a^j)$
 $\mathbf{a}_j = \text{Normalize}(\mathbf{W}(\mu_a^j - \mu_n))$
end for
 $A = [\mathbf{a}_1, \dots, \mathbf{a}_M]$
 $X_{\text{cen}} = \text{Normalize}(X_v) - \mu_n$
 $L = X_{\text{cen}} A$
 $\alpha = \text{Softmax}(L / \tau_{\text{sem}})$
 $S = \sum_{j=1}^M \alpha[:, j] L[:, j]$
 $S_{\text{final}} = \text{GaussianFilter}(S, \sigma)$
Return S_{final}

The text-side covariance is then estimated and the regularized whitening matrix is computed by:

$$\Sigma_T = \text{Cov}(T), \mathbf{W} = (\Sigma_T + \lambda I)^{-1/2}. \quad (12)$$

After obtaining $\mathbf{W}$, PRISM constructs anomaly-specific semantic axes. For the j -th anomaly group, let $\mu_a^j = \text{Mean}(T_a^j)$. The raw anomaly direction is given by the difference between the anomaly center and the normal center, $\mu_a^j - \mu_n$. We then apply the whitening matrix to this direction and normalize it:

$$\mathbf{a}_j = \frac{\mathbf{W}(\mu_a^j - \mu_n)}{\|\mathbf{W}(\mu_a^j - \mu_n)\|_2}. \quad (13)$$

This produces a set of whitened semantic axes

$$A = [\mathbf{a}_1, \dots, \mathbf{a}_M]. \quad (14)$$

For a video embedding v_t , PRISM applies the same L2 normalization and normal-mean centering used in the implementation:

$$\bar{v}_t = \text{Normalize}(v_t) - \mu_n. \quad (15)$$

The alignment logits over all anomaly axes are computed as

$$L_t = \bar{v}_t^\top A. \quad (16)$$

We then aggregate the multi-axis logits with a temperature-controlled semantic weighting:

$$\alpha_t = \text{Softmax}(L_t / \tau_{\text{sem}}), \quad s(v_t) = \sum_{j=1}^M \alpha_{t,j} L_{t,j}. \quad (17)$$

Finally, the frame-level scores are smoothed by a Gaussian filter to obtain S_{final} .

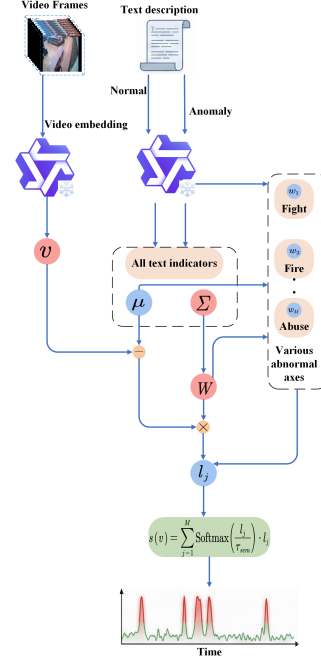


Figure 3. Overall architecture of the PRISM framework. The pipeline extracts visual and textual features via frozen embedding encoders. Text-side mean and covariance statistics are computed from normal and anomaly descriptions to obtain the whitening matrix $\mathbf{W}$. PRISM applies $\mathbf{W}$ to text-derived anomaly directions to construct whitened semantic axes. Video embeddings are L2-normalized and centered, then scored by their alignment with these axes. The final score $s(v)$ is generated via probability-weighted aggregation of alignment logits.

3.3. Overall Implementation Pipeline

The PRISM framework calculates anomaly scores in three stages. First, it embeds normal and anomalous text descriptions and estimates text-side statistics, including the normal semantic center μ , the text covariance Σ , and the whitening matrix $\mathbf{W}$. Second, PRISM applies $\mathbf{W}$ only to the text-derived anomaly directions to construct decorrelated semantic axes. Video embeddings are L2-normalized and centered by μ , but are not whitened by $\mathbf{W}$. Finally, the centered video embeddings are scored by their alignment with the whitened anomaly axes $\{\mathbf{w}_j\}$, followed by probability-weighted semantic aggregation and temporal smoothing. The specific visualization process is shown in Figure 3 and pseudocode Algorithm 1.

4. Experiments

We validated the performance of PRISM using three datasets. In our practical design, we compared it with state-of-the-art VAD methods employing different types of supervision, as well as a model without a training baseline. Furthermore, to verify the necessity of introducing a whitening matrix, we conducted detailed ablation experiments.

Datasets. We use three commonly used VAD datasets to

Table 1. Comparison with state-of-the-art VAD methods. Methods are grouped by category (Unsupervised, Weakly Supervised, Fine-tuning, and Training-free) and sorted by UCF-Crime AUC in ascending order within each group.

Method	Category	UCF-Crime (AUC %)	XD-Violence (AUC %)	XD-Violence (AP %)
Hasan et al.	Unsupervised	51.20	-	50.32
GODS	Unsupervised	70.46	-	61.56
GCL	Unsupervised	74.20	-	-
DyAnNet	Unsupervised	79.76	-	-
Sultani et al.	Weakly Supervised	77.92	-	73.20
SSRL	Weakly Supervised	87.43	-	-
CLIP-TSA(Joo et al., 2023)	Weakly Supervised	87.58	-	82.17
VadCLIP(Wu et al., 2023)	Weakly Supervised	88.02	70.23	84.51
Ex-VAD(Huang et al., 2025)	Weakly Supervised	88.02	70.23	84.51
DSANet(Yin et al., 2025)	Weakly Supervised	89.44	-	86.95
π -VAD(Majhi et al., 2025)	Weakly Supervised	90.33	-	85.37
GS-MoE(D’Amicantonio et al., 2025)	Weakly Supervised	91.58	94.52	82.89
VADor(Öztürk & Can, 2023)	Fine-tuning	85.90	-	-
VERA(Ye et al., 2025)	Fine-tuning	86.55	88.26	70.54
Holmes-VAU(Zhang et al., 2025)	Fine-tuning	88.96	-	87.68
Holmes-VAD(Zhang et al., 2025)	Fine-tuning	89.51	-	90.67
LLaVA-1.5(Liu et al., 2023)	Training-free	72.84	79.62	50.26
Video-Llama2(Cheng et al., 2024)	Training-free	74.42	80.21	53.57
LAVAD(Zanella et al., 2024)	Training-free	80.28	85.36	62.01
EventVAD(Shao et al., 2025)	Training-free	82.03	87.51	64.04
MCANet(Dev et al., 2024)	Training-free	82.47	87.43	69.72
SUVAD(Gao et al., 2025)	Training-free	83.90	-	70.10
VADTree(Li et al., 2025)	Training-free	84.74	90.55	68.85
Flashback(Lee et al., 2025)	Training-free	87.29	90.54	75.13
PRISM (PE)	Training-free	85.85	94.09	80.31
PRISM (Qwen)	Training-free	88.85	93.42	77.48

evaluate our method, which contain real-world surveillance scenarios, namely UCF-Crime(Sultani et al., 2019) , XD-Violence(Wu et al., 2020) and MSAD(Zhu et al., 2024). **UCF-Crime** (Sultani et al., 2019) is a large-scale data set that contains 1900 surveillance videos covering 13 anomalous situations in the real-world. The training set contains 800 normal videos and 810 anomalous videos, while the test set contains 150 normal videos and 140 anomalous videos.

XD-Violence (Wu et al., 2020) is another large-scale dataset for large-scale violence detection, containing 4754 videos with audio signals and weak labels, sourced from movies and YouTube. XD-Violence captures 6 anomalies and splits them into a training set of 3954 videos and a test set of 800 videos.

MSAD (Zhu et al., 2024) is a recently proposed benchmark specifically designed for multi-scenario anomaly detection. Unlike previous datasets limited to single scenes, MSAD collects 720 videos from 14 distinct real-world scenarios (e.g., highways, malls, factories, and parks). It categorizes anomalies into 11 main types (such as fighting, traffic accidents). This benchmark includes 480 training samples and 240 test samples.

Performance Metrics We use the area under the frame-level receiver operating characteristic (ROC) curve (AUC) to mea-

sure the performance of VAD, because AUC is independent of the threshold of the detection task. For the XD-Violence dataset, we also report the Average Precision (AP), which is the area under the frame-level precision-recall curve.

Implementation Details

We employ Qwen3-VL-Embedding-2B(Li et al., 2026) and Perception Encoder-L14(Bolya et al., 2025) as the core multimodal encoder to extract visual and textual features. For video processing, we sampled every 16 frames and used a 7-frame sliding window with a stride of 1 to capture temporal dynamics (reducing the window size for undersampled data). The text used for statistical analysis was consistent with the dataset classification system, containing 20 descriptive sentences per category to ensure semantic diversity. We set a shrinkage regularization term $\lambda = 0.01$ to ensure the numerical stability of the whitening matrix W . Throughout all experiments, the semantic alignment temperature τ_{sem} was set to 0.2 to maintain consistency in the scoring logic. To maintain temporal consistency, we applied a one-dimensional Gaussian filter in post-processing with a smoothing kernel parameter of $\sigma = 16$.

All experiments were performed on a single NVIDIA RTX 4090 GPU. Since this computation added almost no additional computational cost (comparative experiments are

in the appendix), PRISM achieved fast real-time inference speeds on this GPU.

Table 2. Performance comparison on MSAD dataset(using Qwen3-VL-Embedding-2B). Methods are grouped by supervision type and sorted by AUC. Our method (PRISM) achieved the state-of-the-art performance among training-free methods.

Method	Supervision	AUC (%)	AUC _c (%)	AP (%)	AP _c (%)
MGFN	Weakly Supervised	84.96	-	-	-
UR-DMU	Weakly Supervised	85.78	67.95	67.35	75.30
RTFM	Weakly Supervised	86.65	-	-	-
TEVAD(Chen et al., 2023)	Weakly Supervised	86.82	-	-	-
GS-MoE(D’Amicantonio et al., 2025)	Weakly Supervised	87.72	69.54	68.26	76.68
π -VAD(Majhi et al., 2025)	Weakly Supervised	88.68	71.25	71.26	77.86
VADTree(Li et al., 2025)	Training-free	89.32	67.85	71.41	75.49
Flashback (Memory)	Training-free	90.94	66.76	69.93	72.06
PRISM (Qwen)	Training-free	91.84	69.58	75.48	76.50

4.1. Comparison with State of the Art

We evaluate PRISM against current state-of-the-art (SOTA) VAD methods on three benchmarks. As shown in Table 1 and Table 2, PRISM consistently achieves strong performance.

MSAD: PRISM achieves 91.84% AUC, surpassing all training-free and most weakly supervised methods. This shows strong generalization through statistical priors without requiring domain-specific training data.

UCF-Crime: PRISM achieves 88.85% AUC, outperforming VADTree by 4.11% and approaching top weakly-supervised results without any video-level labels.

XD-Violence: Our method attains 77.48% AP, a 8.63% absolute improvement over VADTree, demonstrating effective common-mode noise suppression in complex cinematic scenes. Moreover, PRISM(PE) further improves AP to 80.31%, indicating that PRISM remains effective across different embedding backbones.

4.2. Ablation study

To verify the effectiveness of the core components (statistical whitening and multi-axis semantic aggregation) in the PRISM framework, we designed a 2×3 orthogonal ablation experiment (Table 3).

Experimental Setup

We define six variants: **M1** (Raw) and **M2** (Flashback(Lee et al., 2025)) based on memory; **M3** (Global Raw) and **M4** (Global Whitened) based on global statistics; and **M5** (Multi-Axis(Raw)) It will activate the affinity of the axis for multiple anomalies. **M6** Our complete design (PRISM). Furthermore, we report the signal-to-noise ratio (SNR) to quantify the separability of the feature space.

In addition to the conventional AUC and AP, we introduce Signal-to-Noise Ratio (SNR) as a physical metric to measure the geometric quality of the feature space. It is defined as

Table 3. Ablation study on UCF-Crime and XD-Violence. We report the core metric (AUC for UCF, AP for XD) and Signal-to-Noise Ratio (SNR). **White?**: Whitening, **Gran.**: Granularity. Best results in **bold**. Experimental results using PE as the encoder can be found in the appendix. Note: Since our framework does not rely on extensive auxiliary databases, the reproduction of Flashback(Lee et al., 2025) (M2) was implemented based on the semantic samples provided in our local configuration files, which to ensure a fair comparison under the same data efficiency constraints. The experimental results are consistent with those in Table 3(d) of Flashback(Lee et al., 2025)

ID	Method	White?	Gran.	UCF-Crime		XD-Violence	
				AUC	SNR	AP	SNR
M1	Memory (Raw)	✗	Inst.	81.00	1.11	71.43	1.26
M2	Mem. (Flash.)	✗	Inst.	82.67	1.31	72.99	1.55
M3	Global (Raw)	✗	Glob.	86.15	1.39	75.26	1.82
M4	Global (White.)	✓	Glob.	87.81	1.50	75.69	1.85
M5	(Multi-Axis)	✗	Scen.	88.47	1.51	77.30	1.93
M6	PRISM (Qwen)	✓	Scen.	88.85	1.59	77.48	1.99

follows:

$$\text{SNR} = \frac{\mu_{\text{anom}} - \mu_{\text{norm}}}{\sigma_{\text{norm}}} \quad (18)$$

Where μ_{anom} and μ_{norm} represent the average scores of anomalous and normal frames output by the model, respectively, and σ_{norm} represents the standard deviation of the scores of normal frames. SNR directly quantifies the model’s ability to “suppress background fluctuations (reduce σ_{norm})” and “enhance anomaly responses (increase $\mu_{\text{anom}} - \mu_{\text{norm}}$)”. A higher SNR implies better separability of the feature space.

Analysis of Statistical Whitening

To verify the effectiveness of the whitening mechanism, we compared M3 and M4 in Table 3. Whitening improved the AUC on the UCF-Crime dataset from 86.15% to 87.81%, and the AP on the XD-Violence dataset from 75.26% to 75.69%, while also improving the signal-to-noise ratio (SNR). These results indicate that text-side whitening helps reduce co-directional interference. A similar overall trend is observed with PerceptionEncoder in Appendix Table 4: while whitening is not uniformly beneficial in every global-axis setting, the complete PRISM design consistently achieves the best overall performance when whitening is combined with scenario-aware semantic axes.

Understanding the AP Gap on XD-Violence

As shown in Table 1, on the XD-Violence dataset, the training-free methods still exhibit a significant gap in Average Precision (AP) compared to weakly supervised and fine-tuned methods. This is primarily due to the high intra-class and scene-level variability of this dataset, which includes various movie scenes, moving shots, crowded backgrounds, and rapid scene transitions.

Since the training-free methods rely on frozen global

semantic embeddings, they cannot learn scene-adaptive foreground-background separation from labeled videos. Therefore, normal but dynamic patterns may receive high outlier scores, resulting in difficult-to-identify false positives. These false positives degrade the precision-recall ranking and ultimately reduce AP. PRISM mitigates this problem through text-side whitening and scene-aware semantic axes, but completely resolving non-linear foreground-background entanglement remains a limitation.

Effectiveness of Scenario-Aware Semantic Axes In Table 3, PRISM(Qwen) exhibits the best overall performance, achieving an AUC of 88.85% and an SNR of 1.59 on the UCF-Crime dataset, and an AP of 77.48% and an SNR of 1.99 on the XD-Violence dataset. These results demonstrate that the scene-aware semantic axis achieves a good balance between generalization ability and specificity.

Importantly, this conclusion applies to all backbone networks: as shown in Appendix Table 4, PRISM(PE) also achieves the best overall results, reaching an AUC of 85.85% and an SNR of 1.34 on the UCF-Crime dataset, and an AP of 80.31% and an SNR of 2.06 on the XD-Violence dataset.

4.3. Qualitative Analysis

To visually demonstrate the effectiveness of PRISM, Figure 4 shows the frame-level anomaly scores from four representative samples from UCF-Crime and XD-Violence. The score curves exhibit excellent stability: the anomaly scores are relatively flat in normal footage, while they rise sharply when anomalies occur (red shaded areas). Notably, even in the dynamic backgrounds of XD-Violence (crowds, moving cameras), PRISM maintains a high signal-to-noise ratio, successfully distinguishing normal activity from genuine violent events.

To verify PRISM’s geometric correction and noise suppression ability, we visualize embedding distributions via t-SNE on XD-Violence (Figure 5) and the similarity matrices between video features and the text descriptions on UCF-Crime (Figure 6). All of the above effects are from the qwen3 encoder. The t-SNE visualization on UCF-Crime and the similarity matrix between XD-Violence video features and text descriptions can be viewed in the appendix.

Figure 5 - Geometric Structure: We perform joint projection of the text video embedding and semantic description to ensure that the spatial distance reflects the cosine contraction. Before whitening (left image), the text embedding collapses into a narrow cone, which is semantically irrelevant, and “normal” is buried in the abnormal cluster—which explains the high false positive rate. After PRISM (right), whitening disperses text clusters, which make “Normal” isolated and “Fighting” centered in its cluster, providing qualitative evidence of improved geometric separation.

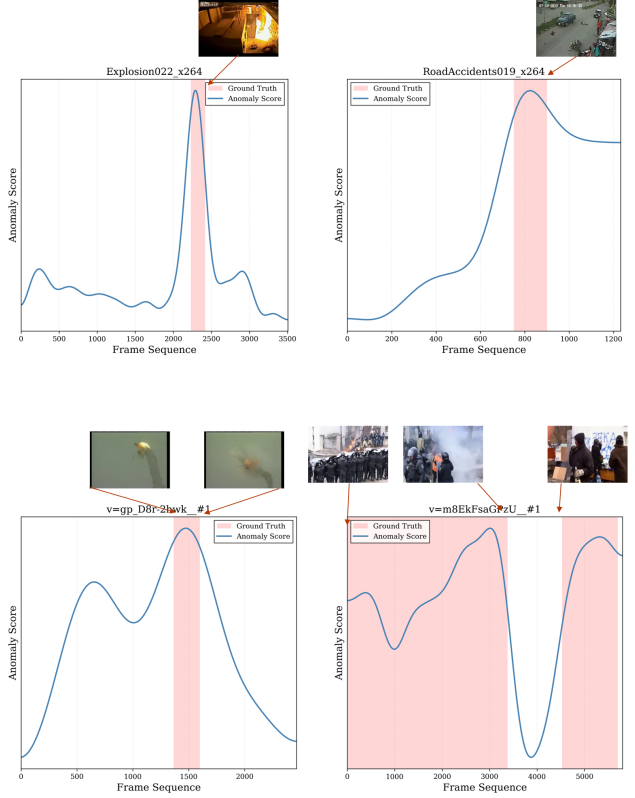


Figure 4. Qualitative results obtained by PRISM on four test videos, including two videos from UCF-Crime (top row) and two videos from XD-Violence (bottom row). For each video, we plotted the inter-frame anomaly scores calculated by PRISM. The figure shows some keyframes and their best-matching temporal summaries (blue lines represent anomaly scores, and red borders represent anomaly frame predictions), visually demonstrating the correlation between predicted anomaly scores and visual content.

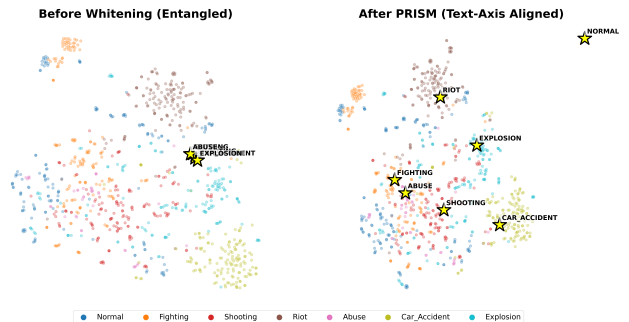


Figure 5. **Visualization of the Joint Feature Space via t-SNE(XD-Violence)** We visualize the geometric distribution of video clips (small dots) and text descriptions (large stars). Different colors represent different events (for example, blue represents an explosion, and red represents a shooting). After applying PRISM, text anchors become better separated from mismatched video clusters, and the normal semantics are more clearly separated from anomalous clusters.

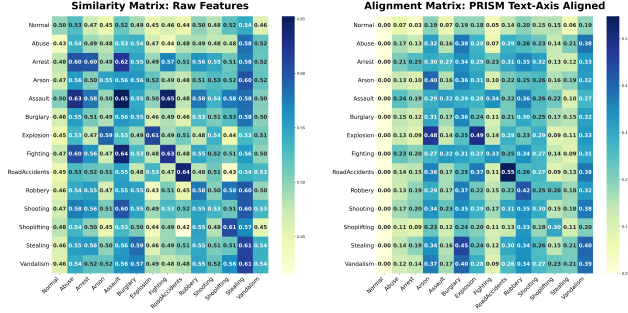


Figure 6. Comparative results of semantic alignment matrix(UCF-Crime). The Y-axis represents the Ground Truth Video Category, and the X-axis represents the text semantic points.

Figure 6 - Quantitative Evidence: Raw features (left) exhibit severe common-mode bias: Normal videos show high similarity to all anomaly texts (0.45-0.53), while diagonal values (correct matches) are only marginally higher (0.52-0.64). After PRISM (right), Normal-Anomaly similarities drop dramatically to near-zero (0.00-0.19), while diagonal structure is preserved. This provides quantitative evidence that PRISM reduces normal-anomaly similarity while preserving semantic alignment.

5. Limitations and Societal Impact

PRISM follows a training-free, text-guided VAD setting and assumes that normal and anomalous semantics can be specified by a predefined description pool. Therefore, PRISM is not a fully open-world anomaly discovery system; detecting completely unspecified anomaly types remains future work. In addition, the current whitening operation is linear and text-side only, which may be insufficient to fully disentangle foreground actions from complex dynamic backgrounds.

6. Conclusion

This paper proposes PRISM, a training-free video anomaly detection framework. PRISM models anomalous biases as common-mode noise caused by anisotropic vector distributions and introduces statistical whitening to eliminate the correlation of background components, thereby significantly improving multiple metrics such as AUC, AP, and signal-to-noise ratio. It also preserves event-specific semantic information. PRISM achieves state-of-the-art performance among training-free methods in three benchmark tests while maintaining real-time inference speed. PRISM requires no training, no auxiliary knowledge base, and no high-latency VLM calls. Together with its interpretable geometric inference process, these properties make PRISM suitable for real-time monitoring and security applications.

Acknowledgements

This work was supported by the AI Project from the Economic and Information Commission of Shanghai, China (Grant No. 2024-GZL-RGZN-01038).

Impact Statement

This research aims to advance efficient and training-free video anomaly detection. Since video anomaly detection systems may be deployed in surveillance and security-critical environments, potential societal risks include privacy issues, false positives, and biased anomaly definitions. Although PRISM does not require storing or reconstructing original video frames, upstream video processing may still involve sensitive visual data. Therefore, we recommend incorporating human-supervised, privacy-preserving data processing measures in practical deployments and conducting careful evaluation under expected operating conditions.

References

- Bolya, D., Huang, P.-Y., Sun, P., Cho, J. H., Madotto, A., Wei, C., Ma, T., Zhi, J., Rajasegaran, J., Bangalath, H., Wang, J., Monteiro, M., Xu, H., Dong, S., Ravi, N., Li, S.-W., Dollar, P., and Feichtenhofer, C. Perception encoder: The best visual embeddings are not at the output of the network. In Belgrave, D., Zhang, C., Lin, H., Pascanu, R., Koniusz, P., Ghassemi, M., and Chen, N. (eds.), *Advances in Neural Information Processing Systems*, volume 38, pp. 60884–60937. Curran Associates, Inc., 2025.
- Chen, W., Ma, K. T., Yew, Z. J., Hur, M., and Khoo, D. A.-A. TEVAD: Improved video anomaly detection with captions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 5549–5559, 2023. doi: 10.1109/CVPRW59228.2023.00587. URL <https://doi.org/10.1109/CVPRW59228.2023.00587>.
- Cheng, Z., Leng, S., Zhang, H., Xin, Y., Li, X., Chen, G., Zhu, Y., Zhang, W., Luo, Z., Zhao, D., and Bing, L. Videollama 2: Advancing spatial-temporal modeling and audio understanding in video-llms, 2024. URL <https://arxiv.org/abs/2406.07476>.
- D’Amicantonio, G., Majhi, S., Kong, Q., Garattoni, L., Francesca, G., Brémond, F., and Bondarev, E. Mixture of Experts guided by Gaussian Splatters Matters: A new approach to weakly-supervised video anomaly detection. *arXiv preprint arXiv:2508.06318*, 2025.
- Dev, P. P., Hazari, R., and Das, P. MCANet: Multimodal caption aware training-free video anomaly detection via large language model. In *Proceedings of the International*

- Conference on Pattern Recognition (ICPR)*, pp. 362–379, 2024.
- Draganov, A., Vadgama, S., Damrich, S., Böhm, J. N., Maes, L., Kobak, D., and Bekkers, E. On the importance of embedding norms in self-supervised learning, 2025. URL <https://arxiv.org/abs/2502.09252>.
- Gao, J., He, D., Tan, X., Qin, T., Wang, L., and Liu, T.-Y. Representation degeneration problem in training natural language generation models. In *International Conference on Machine Learning (ICML)*, pp. 2048–2057, 2019.
- Gao, S., Yang, P., and Huang, L. SUVAD: Semantic understanding based video anomaly detection using MLLM. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1–5, 2025.
- Gong, D., Liu, L., Le, V., Saha, B., Mansour, M. R., Venkatesh, S., and van den Hengel, A. Memorizing normality to detect anomaly: Memory-augmented deep autoencoder for unsupervised anomaly detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 1705–1714, 2019. doi: 10.1109/ICCV.2019.00179. URL <https://doi.org/10.1109/ICCV.2019.00179>.
- Hasan, M., Choi, J., Neumann, J., Roy-Chowdhury, A. K., and Davis, L. S. Learning temporal regularity in video sequences, 2016. URL <https://arxiv.org/abs/1604.04574>.
- Huang, C., Shi, Y., Wen, J., Wang, W., Xu, Y., and Cao, X. Ex-vad: Explainable fine-grained video anomaly detection based on visual-language models. In *Forty-second International Conference on Machine Learning, ICML 2025, Vancouver, BC, Canada, July 13-19, 2025*. OpenReview.net, 2025. URL <https://openreview.net/forum?id=xAhUoyb5eU>.
- Joo, H. K., Vo, K., Yamazaki, K., and Le, N. CLIP-TSA: Clip-assisted temporal self-attention for weakly-supervised video anomaly detection. In *Proceedings of the IEEE International Conference on Image Processing (ICIP)*, pp. 3230–3234, 2023.
- Lee, H., Kim, H., Kim, I.-J., and Choi, Y. Flashback: Memory-driven zero-shot, real-time video anomaly detection, 2025. URL <https://arxiv.org/abs/2505.15205>.
- Li, M., Zhang, Y., Long, D., Chen, K., Song, S., Bai, S., Yang, Z., Xie, P., Yang, A., Liu, D., Zhou, J., and Lin, J. Qwen3-vl-embedding and qwen3-vl-reranker: A unified framework for state-of-the-art multimodal retrieval and ranking, 2026. URL <https://arxiv.org/abs/2601.04720>.
- Li, W., Xu, Y., Rao, Y., Wang, Z., and Deng, S. Vadtree: Explainable training-free video anomaly detection via hierarchical granularity-aware tree, 2025. URL <https://arxiv.org/abs/2510.22693>.
- Liang, Y., Cao, R., Zheng, J., Ren, J., and Gao, L. Learning to remove: Towards isotropic pre-trained bert embedding, 2021. URL <https://arxiv.org/abs/2104.05274>.
- Liu, H., Li, C., Li, Y., and Lee, Y. J. Improved baselines with visual instruction tuning. *arXiv preprint arXiv:2310.03744*, 2023. URL <https://arxiv.org/abs/2310.03744>.
- Liu, W., Luo, W., Lian, D., and Gao, S. Future frame prediction for anomaly detection – a new baseline. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 6536–6545, 2018. doi: 10.1109/CVPR.2018.00684. URL http://openaccess.thecvf.com/content_cvpr_2018/html/Liu_Future_Frame_Prediction_CVPR_2018_paper.html.
- Majhi, S., D’Amicantonio, G., Dantcheva, A., Kong, Q., Garattoni, L., Francesca, G., Bondarev, E., and Brémont, F. Just dance with pi! A poly-modal inductor for weakly-supervised video anomaly detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 24265–24274, 2025.
- Park, H., Noh, J., and Ham, B. Learning memory-guided normality for anomaly detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 14360–14369, 2020. doi: 10.1109/CVPR42600.2020.01438. URL https://openaccess.thecvf.com/content_CVPR_2020/html/Park_Learning_Memory-Guided_Normality_for_Anomaly_Detection_CVPR_2020_paper.html.
- Schakel, A. M. J. and Wilson, B. J. Measuring word significance using distributed representations of words, 2015. URL <https://arxiv.org/abs/1508.02297>.
- Shao, Y., He, H., Li, S., Chen, S., Long, X., Zeng, F., Fan, Y., Zhang, M., Yan, Z., Ma, A., Wang, X., Tang, H., Wang, Y., and Li, S. EventVAD: Training-free event-aware video anomaly detection. In *Proceedings of the ACM International Conference on Multimedia (ACM MM)*, pp. 2586–2595, 2025.
- Sultani, W., Chen, C., and Shah, M. Real-world anomaly detection in surveillance videos, 2019. URL <https://arxiv.org/abs/1801.04264>.

- Wan, B., Fang, Y., Xia, X., and Mei, J. Weakly supervised video anomaly detection via center-guided discriminative learning, 2021. URL <https://arxiv.org/abs/2104.07268>.
- Wu, P., Liu, J., Shi, Y., Sun, Y., Shao, F., Wu, Z., and Yang, Z. Not only look, but also listen: Learning multimodal violence detection under weak supervision, 2020. URL <https://arxiv.org/abs/2007.04687>.
- Wu, P., Zhou, X., Pang, G., et al. VadCLIP: Adapting vision-language models for weakly supervised video anomaly detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.
- Wu, P., Zhou, X., Pang, G., Yang, Z., Yan, Q., Wang, P., and Zhang, Y. Weakly supervised video anomaly detection and localization with spatio-temporal prompts, 2024. URL <https://arxiv.org/abs/2408.05905>.
- Yang, Z., Dai, Z., Salakhutdinov, R., and Cohen, W. W. Breaking the softmax bottleneck: A high-rank RNN language model. In *International Conference on Learning Representations (ICLR)*, 2018. URL <https://openreview.net/forum?id=HkwZsg-0b>.
- Ye, M., Liu, W., and He, P. VERA: explainable video anomaly detection via verbalized learning of vision-language models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025, Nashville, TN, USA, June 11-15, 2025*, pp. 8679–8688. Computer Vision Foundation / IEEE, 2025. doi: 10.1109/CVPR52734.2025.00811. URL https://openaccess.thecvf.com/content/CVPR2025/html/Ye_VERA_Explainable_Video_Anomaly_Detection_via_Verbalized_Learning_of_Vision-Language_CVPR_2025_paper.html.
- Yin, W., Zhang, H., Wang, X., Lu, Y., Zhang, Y., Gong, B., Zuo, J., Yu, L., Gao, C., and Sang, N. Learning to tell apart: Weakly supervised video anomaly detection via disentangled semantic alignment. *arXiv preprint arXiv:2511.10334*, 2025.
- Zanella, L., Menapace, W., Mancini, M., Wang, Y., and Ricci, E. Harnessing large language models for training-free video anomaly detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 18527–18536, 2024.
- Zhang, H., Xu, X., Wang, X., Zuo, J., Huang, X., Gao, C., Zhang, S., Yu, L., and Sang, N. Holmes-VAU: Towards long-term video anomaly understanding at any granularity. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 13843–13853, 2025. doi: 10.1109/CVPR52734.2025.01292.
- Zhu, L., Wang, L., Raj, A., Gedeon, T., and Chen, C. Advancing video anomaly detection: A concise review and a new dataset, 2024. URL <https://arxiv.org/abs/2402.04857>.
- Öztürk, H. I. and Can, A. B. Vador: Real world video anomaly detection with object relations and action. In *34th British Machine Vision Conference 2022, BMVC 2022, Aberdeen, UK, November 20-24, 2023*, pp. 893–898. BMVA Press, 2023. URL <http://proceedings.bmvc2023.org/893/>.

A. appendix.

Table 4. Ablation study on UCF-Crime and XD-Violence using Perception Encoder. PRISM(PE) outperforms the multi-axis baseline and achieves the best overall performance.

ID	Method	White?	Gran.	UCF-Crime		XD-Violence	
				AUC	SNR	AP	SNR
M1	Memory (Raw)	✗	Inst.	81.21	1.10	70.10	1.29
M2	Mem. (Flash.)	✗	Inst.	82.69	1.29	71.89	1.44
M3	Global (Raw)	✗	Glob.	82.01	1.19	77.01	1.91
M4	Global (White.)	✓	Glob.	82.98	1.22	76.32	1.89
M5	(Multi-Axis)	✗	Scen.	85.11	1.30	79.32	2.02
M6	PRISM (PE)	✓	Scen.	85.85	1.34	80.31	2.06

Table 5. End-to-end inference latency and system frame rate (FPS) breakdown across different model backbones.

Method	Video decoding	Image preprocessing	Feature extraction	Algorithm time	Frame rate of end-to-end system
LAVAD	-	-	-	-	1.32 fps
PerceptionEncoder+Flashback	3.56 ms	1.389 ms	14.752 ms	0.410 ms	50.74 fps
Qwen3-VL-Embedding+Flashback	3.96 ms	0.10 ms	86.955 ms	0.532 ms	11.47 fps
PerceptionEncoder+PRISM	3.56 ms	1.389 ms	14.752 ms	0.084 ms	50.75 fps
Qwen3-VL-Embedding+PRISM	3.96 ms	0.10 ms	86.955 ms	0.082 ms	11.48 fps

Table 6. Robustness test on MSAD using mixed text descriptions from UCF-Crime, XD-Violence, and MSAD. Results are based on Qwen3 embeddings.

ID	Method	White?	Gran.	AUC	AUC_A	AP	AP_A	SNR
M1	Memory (Raw)	✗	Inst.	91.10	67.68	69.52	71.90	1.885
M2	Memory (Flash.)	✗	Inst.	91.00	66.38	68.52	70.56	1.943
M3	Global (Raw)	✗	Glob.	91.16	66.82	71.17	72.23	1.934
M4	Global (White.)	✓	Glob.	91.24	67.21	71.08	72.13	1.944
M5	Multi-Axis	✗	Scen.	91.64	68.75	73.49	74.55	1.999
M6	PRISM (Qwen)	✓	Scen.	91.80	69.64	75.18	76.25	2.083

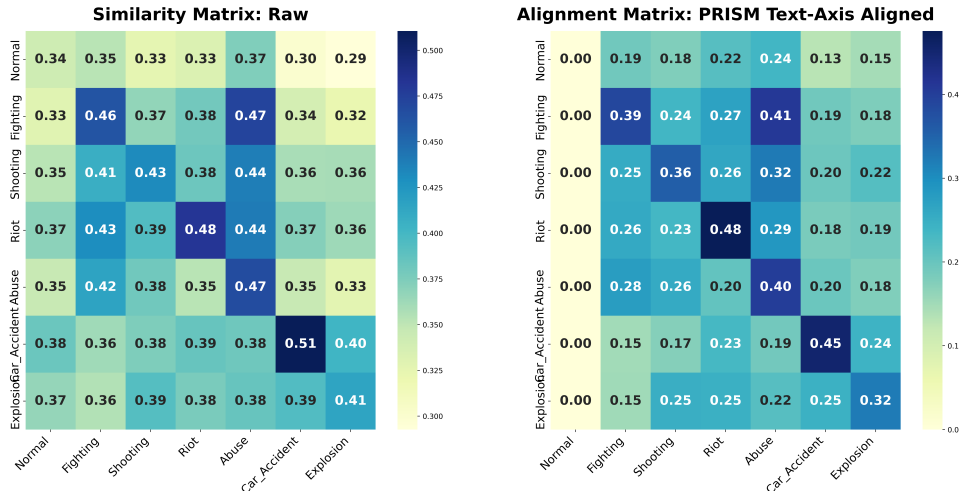


Figure 7. Comparative results of the semantic alignment matrix on XD-Violence.

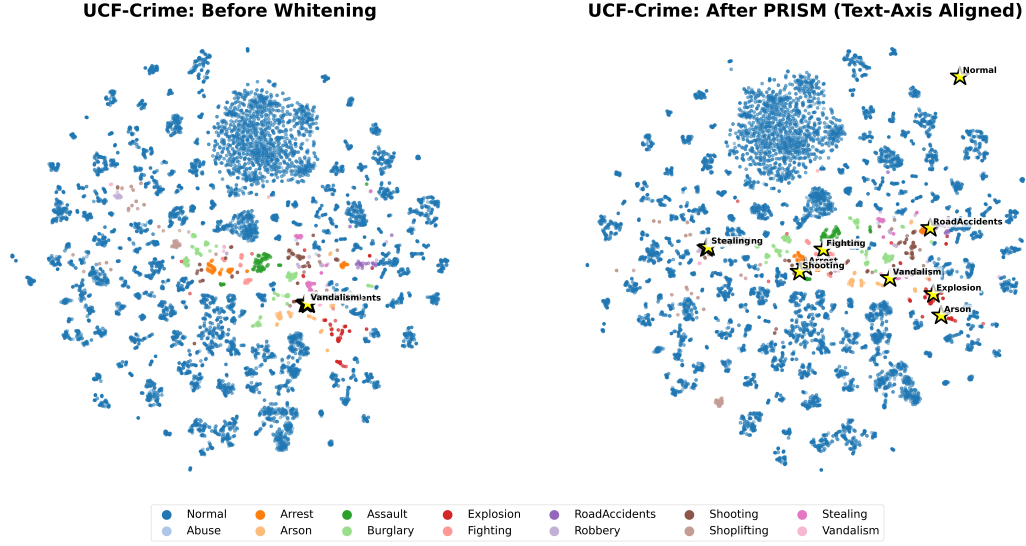


Figure 8. Visualization of the Joint Feature Space via t-SNE (UCF-Crime). Video clips are shown as small dots and text descriptions as large stars, demonstrating robust separation across datasets and conditions.

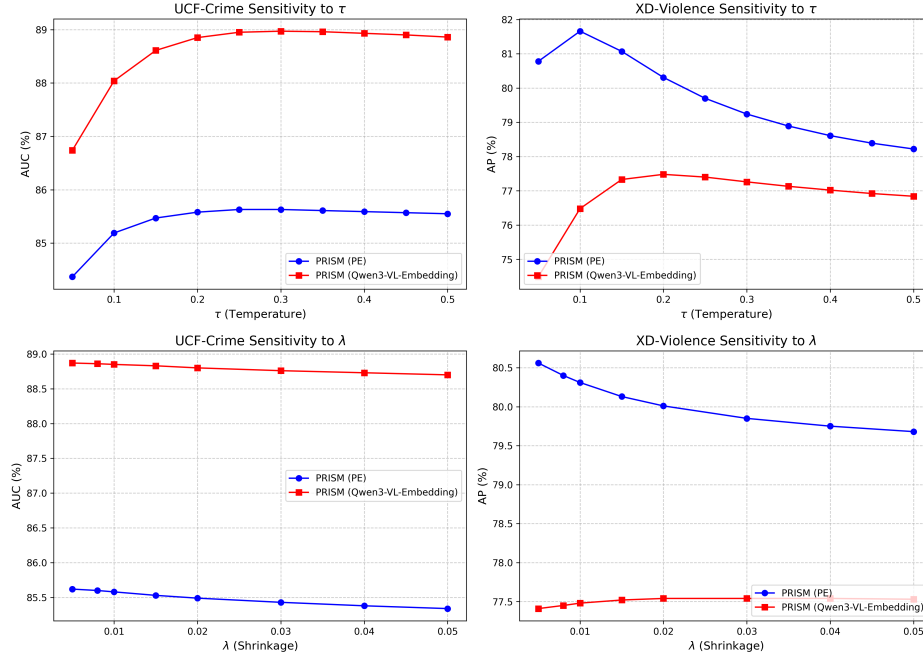


Figure 9. Hyperparameter sensitivity analysis of PRISM. PRISM remains competitive when varying the whitening regularization coefficient λ and semantic temperature τ .